

Medvale Internal Medicine

Chapter 2I: Acute Respiratory Failure, ARDS, Mechanical Ventilation, Hemoptysis, Aspiration, and Miscellaneous Pulmonary Emergencies

This chapter closes the pulmonary section by linking symptom-based emergencies with ICU physiology. The goal is not to memorize devices or ventilator numbers in isolation, but to understand how airway protection, gas exchange, vascular physiology, infection, aspiration, and parenchymal injury interact in real inpatient medicine. A patient with pneumonia can aspirate, develop ARDS, require mechanical ventilation, experience hemoptysis from necrotizing infection, and then develop ventilator-associated complications. These topics therefore belong together as a problem-solving chapter.

Graph-RAG intent: This chapter deliberately repeats mechanistic relationships using clinical language, numeric thresholds, and named associations so nodes can be extracted for causes, findings, diagnostic tests, management steps, complications, and pitfalls.

1. Why This Matters Clinically

Acute respiratory failure is one of the commonest reasons adults deteriorate in the emergency department, ward, or ICU. It is a syndrome, not a diagnosis. The clinician must decide whether the immediate problem is oxygenation, ventilation, airway protection, work of breathing, circulatory shock, or some combination. Hypoxemic respiratory failure is usually a problem of alveolar filling, shunt, severe ventilation-perfusion mismatch, diffusion limitation, or low inspired oxygen. Hypercapnic respiratory failure is usually a problem of inadequate alveolar ventilation from airway obstruction, neuromuscular weakness, chest wall restriction, reduced respiratory drive, or ventilatory muscle fatigue.

The first minutes of care are organized around safety rather than etiologic perfection: assess mental status, airway reflexes, respiratory rate, oxygen saturation, blood pressure, accessory muscle use, ability to speak, and the trajectory of deterioration. A patient who is hypoxemic but alert may be helped by high-flow nasal cannula, noninvasive ventilation, or controlled oxygen. A patient who is obtunded, vomiting, unable to clear secretions, or in shock may need intubation even before the exact pulmonary diagnosis is known.

2. Core Illness Scripts

Syndrome	Typical trigger	Key physiology	Clinical clues	Immediate management frame
Acute hypoxemic respiratory failure	Pneumonia, ARDS, PE, pulmonary edema, aspiration	Low PaO ₂ from shunt or V/Q mismatch; PaCO ₂ may be low early from tachypnea	Tachypnea, low SpO ₂ , diffuse or focal infiltrates, increased work of breathing	Oxygen escalation, identify reversible cause, consider HFNC/NIV/intubation
Acute hypercapnic respiratory failure	COPD/asthma exacerbation, sedatives/opioids, neuromuscular weakness	Alveolar ventilation is inadequate; PaCO ₂ rises and pH falls	Somnolence, headache, asterixis, wheeze or weak cough, acidemia	Reverse cause, NIV when appropriate, intubate if failing
ARDS	Sepsis, pneumonia, aspiration, pancreatitis, trauma, transfusion	Diffuse inflammatory alveolar-capillary injury causing noncardiogenic edema and shunt	Bilateral opacities, severe hypoxemia despite oxygen, low compliance	Lung-protective ventilation, PEEP, prone positioning for severe cases
Massive/life-threatening hemoptysis	Bronchiectasis, TB, cancer, necrotizing infection, vascular lesion	Asphyxiation risk from blood flooding airways more than exsanguination	Coughing blood, hypoxemia, airway obstruction, unilateral source may be present	Airway first, bleeding side down, CT/CTA/bronchoscopy, bronchial artery embolization
Aspiration event	Dysphagia, seizure, intoxication, anesthesia, vomiting, impaired cough	Chemical injury or bacterial inoculation of dependent lung segments	Acute cough/wheeze/hypoxemia after event or delayed fever/infiltrate	Supportive care for pneumonitis; antibiotics for pneumonia or complications

3. Acute Respiratory Failure: A Practical Framework

Respiratory failure is traditionally divided into type 1 hypoxemic failure and type 2 hypercapnic failure, but real patients frequently have overlap. A patient with COPD and pneumonia may have both low PaO₂ from V/Q mismatch and high

PaCO₂ from airway obstruction and fatigue. A patient with ARDS may initially hyperventilate with a low PaCO₂, then later develop hypercapnia as dead space increases or ventilatory muscles fail. A graph representation should therefore allow respiratory failure to connect to both gas-exchange abnormalities and mechanical pump failure.

Numeric anchors are useful. A PaO₂ below about 60 mm Hg or oxygen saturation below about 90% usually signals clinically important hypoxemia. A PaCO₂ above 45 mm Hg indicates hypercapnia, but the pH determines acuity: chronic CO₂ retainers may have bicarbonate compensation and near-normal pH, whereas acute ventilatory failure causes respiratory acidosis. A rapidly rising PaCO₂, worsening acidemia, altered mental status, exhaustion, or inability to protect the airway should lower the threshold for intubation.

Problem	ABG pattern	Common causes	Treatment target
Pure hypoxemia	Low PaO ₂ ; PaCO ₂ often low or normal early	Pneumonia, ARDS, PE, pulmonary edema, atelectasis	Increase alveolar oxygen delivery, recruit alveoli, treat cause
Ventilatory failure	High PaCO ₂ ; pH low if acute	COPD/asthma, opioid toxicity, neuromuscular disease, obesity hypoventilation	Increase alveolar ventilation and reduce work of breathing
Shunt physiology	Hypoxemia poorly responsive to oxygen	ARDS, lobar pneumonia, atelectasis, intracardiac shunt	PEEP/recruitment, treat alveolar collapse or filling
Dead-space physiology	High minute ventilation need; CO ₂ may rise	PE, emphysema, shock, overdistension on ventilator	Improve perfusion, avoid excessive overdistension, treat clot/shock
Airway protection failure	Gas numbers may be normal until sudden decompensation	Stroke, intoxication, seizure, vomiting, severe weakness	Intubation when aspiration or obstruction risk is high

4. ARDS: Diffuse Alveolar Injury and Refractory Hypoxemia

Acute respiratory distress syndrome is a final common pathway of severe inflammatory lung injury. The initiating insult may be direct, such as pneumonia, aspiration, inhalation injury, or pulmonary contusion, or indirect, such as sepsis, pancreatitis, trauma, transfusion, or shock. In either route, endothelial and epithelial injury increases alveolar-capillary permeability. Protein-rich edema enters alveoli, surfactant function worsens, alveoli collapse, and the lung becomes stiff. The patient then develops hypoxemia that often responds incompletely to supplemental oxygen because a substantial fraction of cardiac output passes through nonventilated or poorly ventilated alveoli.

The Berlin framework defines ARDS by acute timing, bilateral opacities on chest imaging, respiratory failure not fully explained by cardiac failure or fluid overload, and oxygenation impairment measured by PaO₂/FiO₂ with at least 5 cm H₂O of PEEP or CPAP. Severity is based on the PaO₂/FiO₂ ratio: mild is >200 to ≤300 mm Hg, moderate is >100 to ≤200 mm Hg, and severe is ≤100 mm Hg. These categories are not merely labels; worsening categories correlate with higher mortality and often trigger escalation such as higher PEEP strategies, prone positioning for severe cases, or extracorporeal support consideration in specialized centers.

Original mechanism diagram: ARDS pathway

Trigger: sepsis, pneumonia, aspiration, trauma	Alveolar-capillary injury and leak	Surfactant dysfunction and collapse	Low compliance + shunt hypoxemia	Low Vt ventilation + PEEP + prone if severe
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ARDS criterion	Operational anchor	Clinical meaning
Timing	Within 1 week of known insult or new/worsening respiratory symptoms	ARDS is acute, not a chronic fibrotic or longstanding edema pattern
Imaging	Bilateral opacities not fully explained by effusions, lobar collapse, or nodules	Both lungs are involved by alveolar filling/collapse/injury
Origin of edema	Not fully explained by heart failure or fluid overload	Echo, BNP, exam, and volume history may be needed to separate from hydrostatic edema
Oxygenation	PaO ₂ /FiO ₂ ≤300 with PEEP/CPAP ≥5 cm H ₂ O	Severity is quantified by oxygenation defect under positive pressure
Severity	Mild >200-300; moderate >100-200; severe ≤100	Severity influences rescue strategies and prognosis

A common exam and clinical pitfall is confusing ARDS with cardiogenic pulmonary edema. Both can cause bilateral infiltrates and hypoxemia. ARDS is favored by a recent inflammatory insult, normal or low filling pressures, poor compliance, and noncardiogenic edema. Cardiogenic pulmonary edema is favored by elevated jugular venous pressure, S₃, severe hypertension or ischemia, cardiomegaly, Kerley lines, pleural effusions, and rapid improvement with diuresis or

afterload reduction. However, overlap is common; septic patients can receive large fluid volumes and develop both capillary leak and hydrostatic edema.

5. Mechanical Ventilation: What the Numbers Mean

Mechanical ventilation is a supportive therapy, not a cure. It buys time while antibiotics, anticoagulation, bronchodilators, diuresis, source control, or immune-directed therapy treat the cause. The ventilator manipulates oxygenation mainly through FiO₂ and PEEP and manipulates ventilation mainly through minute ventilation, which is tidal volume multiplied by respiratory rate. The clinician must balance gas exchange against lung injury: higher pressures and volumes may improve numbers temporarily while worsening barotrauma, volutrauma, atelectrauma, hemodynamics, and inflammation.

Lung-protective ventilation in ARDS uses a low tidal volume strategy, commonly about 6 mL/kg predicted body weight rather than actual body weight, because lung size tracks height and sex more than obesity. Plateau pressure is generally targeted at ≤ 30 cm H₂O, and driving pressure is often followed as a marker of stress on the respiratory system. Permissive hypercapnia means accepting a higher PaCO₂ if the pH is tolerable in order to avoid injurious ventilation; this is generally avoided or used cautiously in situations where hypercapnia is dangerous, such as severe intracranial hypertension.

Setting/measure	Primary effect	Useful numeric anchor	Pitfall
FiO ₂	Raises inspired oxygen fraction	Start high in emergency; reduce when stable to avoid oxygen toxicity	A high FiO ₂ alone does not recruit collapsed alveoli
PEEP	Keeps alveoli open at end-expiration	ARDS requires at least 5 cm H ₂ O for Berlin oxygenation categories	Too much PEEP can overdistend alveoli, lower venous return, and raise dead space
Tidal volume	Controls breath size and contributes to ventilation	ARDS target often ~ 6 mL/kg predicted body weight	Using actual body weight in obesity can overventilate the lung
Respiratory rate	Controls minute ventilation with tidal volume	Raise rate to lower PaCO ₂ , within limits	Very high rates can cause air trapping, especially in obstruction
Plateau pressure	Estimates alveolar distending pressure during no flow	Common ARDS target ≤ 30 cm H ₂ O	High plateau suggests poor compliance or excessive tidal volume
Peak pressure	Reflects airway resistance plus elastic pressure	High peak with normal plateau suggests resistance/mucus/bronchospasm	Do not confuse high peak alone with alveolar overdistension
Pressure support/SBT	Assesses ability to breathe with assistance reduced	Used during weaning when cause is improving	Passing an SBT does not guarantee airway protection

Oxygenation failure on a ventilator can be approached by checking the patient before changing settings: confirm endotracheal tube position, suction mucus, assess breath sounds, evaluate for pneumothorax, review hemodynamics, inspect the circuit, and compare chest imaging. Ventilator changes are then chosen by physiology. Low saturation with recruitable atelectasis may improve with PEEP. Low saturation from mucus plugging may require suction or bronchoscopy. Low saturation from shock may improve only when perfusion improves. Hypercapnia from severe obstruction may worsen if the respiratory rate is increased so much that the patient cannot exhale.

6. Noninvasive Ventilation, High-Flow Oxygen, and Intubation

High-flow nasal cannula provides heated humidified oxygen at high flows, reduces entrainment of room air, can generate a small amount of positive airway pressure, and decreases work of breathing. It is useful in many forms of acute hypoxemic respiratory failure when the patient can protect the airway and is not in immediate arrest. Noninvasive ventilation provides positive pressure through a mask and is especially useful for COPD exacerbation with acute hypercapnic acidosis and for cardiogenic pulmonary edema. It can fail when secretions are heavy, mental status is impaired, facial seal is poor, shock is present, or hypoxemia is severe and progressive.

Intubation is not defined by a single oxygen saturation. It is a decision about trajectory and safety. Indications include inability to protect the airway, persistent or worsening hypoxemia despite appropriate support, severe work of breathing with fatigue, worsening hypercapnic acidosis, hemodynamic instability, recurrent aspiration, or need for controlled ventilation during procedures or shock. In severe asthma or COPD, intubation is high risk because dynamic hyperinflation and auto-PEEP can cause hypotension and barotrauma; lower respiratory rates and longer expiratory times are often needed.

Modality	Best fit	Avoid or use cautiously when	Key failure clues
Conventional oxygen	Mild hypoxemia with stable work	Rapid deterioration or high	Increasing flow need, tachypnea,

	of breathing	oxygen requirement	confusion
High-flow nasal cannula	Hypoxemic failure with intact airway reflexes	Immediate need for airway protection or severe shock	Persistent hypoxemia, rising work of breathing, exhaustion
NIV/BiPAP	COPD exacerbation with acidosis, cardiogenic pulmonary edema, selected OHS	Vomiting, copious secretions, altered mental status, facial trauma, inability to cooperate	Worsening pH/PaCO ₂ , persistent tachypnea, mask intolerance, aspiration
Invasive ventilation	Airway protection failure, severe ARDS, shock, failing NIV/HFNC	No absolute avoidance when lifesaving	Hypotension after intubation, pneumothorax, dyssynchrony, ventilator injury

7. Hemoptysis: Blood in the Airway Is an Airway Problem

Hemoptysis is expectoration of blood from the lower respiratory tract. It must be separated from pseudohemoptysis, in which blood originates from the nose, mouth, or throat, and from hematemesis, in which blood originates from the gastrointestinal tract. Clinically important hemoptysis ranges from blood-streaked sputum to life-threatening airway flooding. Traditional volume categories include minor hemoptysis below about 30 mL, moderate or major hemoptysis around 30-300 mL, and massive hemoptysis often described as more than 300-400 mL in 24 hours, but the modern bedside concept is functional: hemoptysis is life-threatening when it causes airway compromise, hypoxemia, hemodynamic instability, or inability to clear blood.

Most severe hemoptysis arises from the bronchial circulation rather than the pulmonary arterial circulation because bronchial arteries are systemic-pressure vessels. Important causes include bronchiectasis, tuberculosis, necrotizing pneumonia, lung abscess, malignancy, pulmonary embolism or infarction, vasculitis such as granulomatosis with polyangiitis, diffuse alveolar hemorrhage, anticoagulation, and iatrogenic airway trauma. In a graph, hemoptysis should therefore connect to infection, structural airway disease, cancer, vascular disease, immune capillaritis, and medication-related bleeding.

Original mechanism diagram: life-threatening hemoptysis pathway

Airway safety? distress, hypoxemia, large volume	Bleeding side down to protect good lung	Localize with CXR + CT/CTA or bronchoscopy	Control with bronchoscopy and/or bronchial artery embolization
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Hemoptysis cause	Clues	Diagnostic direction	Management idea
Bronchiectasis	Chronic productive cough, recurrent infections, prior CT airway dilation	CT chest/CTA; sputum cultures if infection	Antibiotics for exacerbation, airway clearance, embolization if severe
Tuberculosis or fungal infection	Fever, night sweats, cavitary lesion, exposure risk, immunosuppression	AFB testing, cultures, CT, isolation when TB possible	Pathogen-directed therapy; public health precautions
Malignancy	Age, smoking, weight loss, recurrent pneumonia, central mass	CT chest with contrast, bronchoscopy/biopsy as indicated	Cancer staging/treatment; bronchoscopic control or embolization for bleeding
Pulmonary embolism/infarct	Pleuritic pain, dyspnea, VTE risk, small-volume blood	CTPA or V/Q depending context	Anticoagulation unless bleeding risk dominates; advanced PE therapy if unstable
Diffuse alveolar hemorrhage	Anemia, bilateral infiltrates, renal disease, autoimmune clues	CBC, urinalysis, serologies, bronchoscopy with progressively bloody lavage	Immunosuppression/plasma exchange in selected etiologies; ICU support
Iatrogenic/anticoagulant	Recent procedure, supratherapeutic anticoagulation, airway trauma	Medication review, coagulation studies, bronchoscopy if needed	Hold/reverse anticoagulation when appropriate; local control

Initial management of life-threatening hemoptysis prioritizes airway protection and preservation of the nonbleeding lung. Sit the patient upright if stable, provide oxygen, obtain large-bore IV access, correct coagulopathy when appropriate, and call pulmonary/interventional radiology early. If one side is known or suspected to be bleeding, place the bleeding side down so gravity reduces contamination of the opposite lung. CT with contrast or CTA is often used to localize the source and plan bronchial artery embolization, while bronchoscopy can localize active bleeding, clear clots, isolate a lung, and provide local therapies. Bronchial artery embolization is a key therapy for many severe or recurrent cases.

8. Aspiration: Pneumonitis, Pneumonia, Abscess, and Prevention

Aspiration means material enters the lower respiratory tract from above. The aspirated material may be acidic gastric contents, food particles, oral secretions, colonized oropharyngeal contents, blood, or water. The clinical syndrome depends on the volume, pH, bacterial burden, host defenses, and baseline lung function. A witnessed large-volume gastric aspiration during intoxication or anesthesia can cause abrupt chemical pneumonitis within minutes to hours. Silent microaspiration in a frail patient with dysphagia can cause recurrent pneumonias over weeks to months.

Original mechanism diagram: aspiration syndromes

Aspiration event: gastric or oral contents	Chemical pneumonitis: minutes-hours	Aspiration pneumonia: infection evolves	Complications: abscess, empyema, ARDS
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Aspiration pneumonitis is a chemical injury. It often presents abruptly after a witnessed event with cough, bronchospasm, fever, leukocytosis, hypoxemia, and infiltrates that can mimic infection. Because inflammation alone can cause fever and leukocytosis, antibiotics are not automatically required if the patient improves rapidly and bacterial infection is not suspected. Aspiration pneumonia is infection caused by aspirated colonized material. It is favored by delayed or persistent fever, purulent sputum, progressive infiltrates, frailty, poor dentition, dysphagia, institutionalization, or lack of improvement after 24-48 hours.

Dependent-lobe anatomy helps localize aspiration. In an upright patient, the basal segments of the lower lobes are common. In a supine patient, the posterior upper lobes and superior segments of the lower lobes are common. Right-sided involvement is frequent because the right main bronchus is more vertical. Lung abscess classically presents with subacute fever, weight loss, foul-smelling sputum, and a cavitary lesion with an air-fluid level; it suggests necrotizing infection and often requires prolonged antibiotics. Empyema requires drainage rather than antibiotics alone.

Feature	Aspiration pneumonitis	Aspiration pneumonia	Lung abscess/empyema
Primary process	Chemical inflammatory injury	Bacterial infection after aspiration	Necrotizing infection or infected pleural space
Timing	Abrupt, minutes-hours after event	Often delayed or persistent after aspiration risk	Subacute or progressive; may follow pneumonia
Antibiotics	Not routine if rapid improvement and no infection evidence	Treat according to CAP/HAP risk and severity	Antibiotics plus drainage when empyema; prolonged therapy for abscess
Anaerobic coverage	Not a default requirement	Selective; consider with abscess, empyema, severe periodontal disease, necrosis	Often relevant, especially abscess/empyema
Prevention	Airway protection, positioning, cautious sedation	Swallow evaluation, oral care, feeding strategy, head-of-bed elevation	Source control, dental care, prevent recurrence

Modern pneumonia guidance does not support routine anaerobic coverage for every suspected aspiration pneumonia. Anaerobic therapy is most appropriate when there is lung abscess, empyema, necrotizing infection, or strong evidence of anaerobic oral source. Aspiration risk alone does not automatically equal clindamycin or broad anaerobic therapy.

9. Miscellaneous Pulmonary Emergencies and Common Problems

Atelectasis is alveolar collapse and is common after surgery, mucus plugging, shallow breathing, or obstruction. It causes low lung volume, basilar opacity, and sometimes fever early postoperatively, but postoperative fever should not be reflexively attributed to atelectasis without assessing infection, thrombosis, drug fever, and surgical complications. Prevention and treatment include mobilization, pain control that permits deep breathing, incentive spirometry, positive pressure when needed, and bronchoscopy for refractory mucus plugging or lobar collapse.

Mucus plugging is especially important in asthma, COPD, bronchiectasis, cystic fibrosis, dehydration, intubation, and weak cough. It can cause abrupt hypoxemia, unilateral wheeze or absent breath sounds, lobar collapse, increased peak airway pressure on a ventilator, and difficulty passing suction catheters. Treatment is airway humidification, bronchodilators when bronchospasm is present, chest physiotherapy, suctioning, and bronchoscopy if obstruction persists.

Inhalation injury should be suspected after enclosed-space fires, facial burns, soot in the mouth or sputum, hoarseness, stridor, carbon monoxide exposure, cyanide toxicity, or altered mental status. Pulse oximetry can be misleading in carbon monoxide poisoning because carboxyhemoglobin may be read as oxyhemoglobin. Treatment priorities are early airway

evaluation, high-flow oxygen, consideration of hyperbaric oxygen for selected severe carbon monoxide poisoning, and cyanide antidote therapy when clinically indicated.

Pulmonary contusion is traumatic alveolar hemorrhage and edema without necessarily having an external wound. It may worsen over the first 24-48 hours after blunt trauma and can precipitate hypoxemia or ARDS. Management is supportive oxygenation, cautious fluids, pain control, pulmonary hygiene, and ventilation when necessary. Flail chest adds mechanical instability and worsens ventilation by increasing work of breathing; analgesia and ventilatory support may be as important as treating the contusion itself.

Problem	Classic trigger	Key clue	Management concept
Atelectasis	Postoperative splinting, mucus plug, obstruction	Volume loss, basilar opacity, low-grade symptoms	Mobilize, analgesia, deep breathing, positive pressure, remove obstruction
Mucus plugging	COPD, asthma, bronchiectasis, intubation	Sudden hypoxemia, lobar collapse, high peak pressure	Humidify, suction, bronchodilate, airway clearance, bronchoscopy
Inhalation injury	Closed-space fire, smoke exposure	Soot, hoarseness, stridor, CO/cyanide toxicity	Airway first, 100% oxygen, early intubation if edema risk
Pulmonary contusion	Blunt chest trauma	Hypoxemia and infiltrate after trauma, may worsen 24-48 h	Support oxygenation, pain control, avoid fluid overload
High-altitude pulmonary edema	Rapid ascent, exertion, prior HAPE	Dyspnea, cough, hypoxemia, crackles at altitude	Descent, oxygen, rest, nifedipine in selected cases
Negative-pressure pulmonary edema	Acute upper airway obstruction, laryngospasm	Pink frothy sputum after obstruction relief	Airway support, oxygen/PEEP, usually rapid improvement

10. Diagnostic Approach to the Crashing Pulmonary Patient

A structured pulmonary emergency evaluation prevents anchoring. First, determine whether the airway is patent and protected. Second, determine whether oxygenation is adequate and whether the patient is tiring. Third, rapidly screen for immediately reversible killers: tension pneumothorax, massive PE, severe bronchospasm, mucus plug, cardiogenic pulmonary edema, sepsis, aspiration, and equipment failure in ventilated patients. Fourth, interpret imaging and blood gases in the context of the trajectory, not as isolated numbers.

Chest x-ray is fast and useful for pneumothorax, lobar pneumonia, pulmonary edema, pleural effusion, tube position, and gross atelectasis. CT gives more detail for PE, hemoptysis source localization, malignancy, occult pneumonia, abscess, complex pleural disease, and trauma. Bedside ultrasound can rapidly detect pleural effusion, pneumothorax signs, B-lines, right ventricular strain, and gross cardiac dysfunction. ABG or VBG can identify acidemia and hypercapnia, while lactate, CBC, chemistry, troponin/BNP, cultures, and ECG clarify systemic severity and competing diagnoses.

Clinical question	Fastest useful data	How it changes management
Is this airway failure?	Mental status, gag/cough, secretions, vomiting, facial/neck trauma	Triggers early intubation or airway specialist help
Is this oxygenation failure?	SpO ₂ , ABG PaO ₂ , CXR/ultrasound/CT	Escalates oxygen, PEEP, HFNC, NIV, intubation, cause-directed therapy
Is this ventilation failure?	ABG/VBG pH and PaCO ₂ , respiratory rate, fatigue signs	NIV for selected patients or invasive ventilation if failing
Is there shock?	BP, lactate, urine output, echo, extremities	Treat sepsis/PE/tension pneumothorax/cardiogenic causes; intubation risk increases
Is there a focal mechanical problem?	Breath sounds, CXR, ultrasound, ventilator pressures	Needle/chest tube, suction/bronchoscopy, tube repositioning
Is infection driving decline?	Fever, WBC, cultures, imaging, source exam	Early antibiotics, source control, sepsis bundle principles

11. Initial Management Algorithms

The initial management sequence for severe pulmonary deterioration can be remembered as A-B-C plus cause. Airway: protect the airway if the patient cannot protect it, and anticipate difficult intubation in angioedema, burns, trauma, obesity, or severe hypoxemia. Breathing: provide oxygen, bronchodilators, PEEP, ventilation, drainage, or decompression depending on physiology. Circulation: treat shock, because hypoxemia and hypotension amplify each other. Cause: give antibiotics for pneumonia/sepsis, anticoagulate PE when safe, drain empyema or pneumothorax, reverse opioid toxicity, treat bronchospasm, or embolize bleeding vessels.

In ARDS, treatment is supportive and cause-directed: treat sepsis or pneumonia, avoid excessive fluids after shock resuscitation, use lung-protective ventilation, maintain adequate PEEP, consider prone positioning for severe hypoxemia, prevent complications such as VAP and venous thromboembolism, and start liberation from the ventilator as soon as the cause improves. In hemoptysis, treatment is airway-directed: protect the nonbleeding lung, localize bleeding, reverse coagulopathy when appropriate, and use bronchoscopy or bronchial artery embolization for ongoing or life-threatening bleeding. In aspiration, treatment is syndrome-directed: supportive care for chemical pneumonitis, antibiotics for bacterial pneumonia, and drainage/source control for abscess or empyema.

12. Complications and Pitfalls

- Calling every post-aspiration infiltrate aspiration pneumonia. Chemical pneumonitis can mimic infection; antibiotics are driven by clinical course and evidence of infection, not the aspiration event alone.
- Waiting for CT before treating tension physiology or airway-flooding hemoptysis. Unstable obstructive shock and life-threatening airway bleeding are bedside emergencies.
- Using actual body weight for ARDS tidal volume. Predicted body weight is based on height and sex because it estimates lung size.
- Increasing respiratory rate in severe obstructive disease without checking expiratory time. This can worsen air trapping, auto-PEEP, hypotension, and barotrauma.
- Assuming pulse oximetry is reliable after smoke inhalation. Carbon monoxide poisoning may produce falsely reassuring readings.
- Assuming hemoptysis volume alone defines danger. A smaller amount can be life-threatening in a patient with poor reserve or inability to clear blood.
- Forgetting that NIV can delay needed intubation. Worsening acidosis, fatigue, shock, aspiration risk, or altered mental status should trigger escalation.

13. High-Yield Exam Clues

Clue	Think of	Why
PaO ₂ /FiO ₂ ≤100 on PEEP ≥5 with bilateral opacities after sepsis	Severe ARDS	Severe shunt physiology from inflammatory alveolar injury
High peak pressure but normal plateau pressure	Airway resistance problem	Bronchospasm, mucus plug, kinked tube, biting tube
High peak and high plateau pressure	Low compliance problem	ARDS, pulmonary edema, pneumothorax, abdominal compartment, stiff chest wall
COPD patient on BiPAP with improving pH and falling PaCO ₂	Successful NIV physiology	Ventilatory pump support is reversing acute hypercapnia
Bright red blood with cough and recurrent infections	Bronchiectasis hemoptysis	Dilated inflamed airways supplied by systemic bronchial arteries can bleed
Witnessed vomiting followed by abrupt wheeze/hypoxemia	Aspiration pneumonitis	Chemical injury occurs quickly after macroaspiration
Foul sputum plus cavitary lesion with air-fluid level	Lung abscess	Necrotizing infection often involves anaerobic/oral flora patterns
Soot in mouth after house fire with hoarseness	Inhalation injury	Airway edema can progress; early airway control may be safer

14. Original Concept Relationships for Graph RAG

Source node	Relationship	Target node	Rationale
ARDS	causes	refractory hypoxemia	Shunt and alveolar collapse make oxygenation difficult even with high FiO ₂
ARDS	managed by	low tidal volume ventilation	Lower stretch reduces ventilator-induced lung injury
PEEP	improves	alveolar recruitment	End-expiratory pressure reduces repetitive collapse
Excessive PEEP	can cause	hypotension	Increased intrathoracic pressure reduces venous return and can overdistend alveoli
Hemoptysis	threatens	airway patency	Death often occurs from asphyxiation rather than blood loss

Bronchial arteries	supply	major hemoptysis source	Systemic-pressure vessels are common bleeding source
Aspiration pneumonitis	differs from	aspiration pneumonia	Chemical injury is not the same as bacterial infection
Lung abscess	suggests	anaerobic or necrotizing infection pattern	Cavitation and foul sputum imply tissue necrosis and source-control thinking
Mucus plugging	causes	atelectasis	Airway obstruction leads to distal alveolar collapse
Smoke inhalation	can cause	carbon monoxide poisoning	Carboxyhemoglobin impairs oxygen delivery and misleads pulse oximetry

15. Practice Questions

Question 1: A 58-year-old man with septic shock from pneumonia is intubated. Chest imaging shows bilateral opacities. On PEEP 8 cm H₂O, PaO₂ is 70 mm Hg on FiO₂ 0.70. What ARDS severity category does this meet?

Answer: The PaO₂/FiO₂ ratio is 100. With PEEP ≥5 and compatible timing/imaging, this is severe ARDS because severe ARDS is PaO₂/FiO₂ ≤100.

Question 2: A ventilated patient suddenly has high peak airway pressure, unchanged plateau pressure, wheezing, and difficulty passing a suction catheter. What physiology is most likely?

Answer: The pattern suggests increased airway resistance rather than low compliance. Bronchospasm, mucus plugging, a kinked tube, or biting the tube should be considered.

Question 3: A patient with bronchiectasis coughs up a large amount of bright red blood and becomes hypoxemic. What is the first management priority?

Answer: Airway protection and preservation of the nonbleeding lung come first. Position bleeding side down if known, provide oxygen, call for pulmonary/interventional radiology help, and evaluate for bronchoscopy or bronchial artery embolization.

Question 4: An intoxicated patient vomits, has abrupt hypoxemia and wheezing, then improves substantially over 12 hours without persistent fever. What syndrome is most consistent?

Answer: Aspiration pneumonitis is most consistent because the time course is abrupt chemical injury after macroaspiration with rapid improvement. Antibiotics are not automatic if bacterial infection does not evolve.

Question 5: An older nursing-home resident with dysphagia develops fever, leukocytosis, purulent sputum, and a dependent right lower-lobe infiltrate 48 hours after a choking episode. What syndrome is most consistent?

Answer: Aspiration pneumonia is more likely because infectious features persist and evolve after aspiration risk. Antibiotic choice should match severity and CAP/HAP risk; anaerobic coverage is selective rather than automatic.

Question 6: After a house fire, a patient has headache and confusion but pulse oximetry is 99%. Why can the oximeter be misleading?

Answer: Carbon monoxide poisoning can cause falsely reassuring pulse oximetry because standard pulse oximetry cannot reliably distinguish oxyhemoglobin from carboxyhemoglobin.

16. References and Clinical Anchors

- The uploaded pulmonary reference was used as a broad coverage and sequencing signal for pulmonary emergencies and miscellaneous topics, not as wording or image source.
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